

On the peridynamic formulation for an orthotropic Mindlin plate under bending

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Abstract

Composite materials are becoming increasingly widespread across applications such as aerospace, automotive, wind and solar energy systems, on account of their superior specific strength and stiffness. There are still unresolved challenges in the structural analysis of composite materials in practical engineering applications, owing to their complex structures. Peridynamic theory is a robust technique for the analysis of composite structures; however, it is computationally costly. Structural idealizations are commonly applied in engineering applications to represent structures with a reasonable simplicity, thus reducing the computational cost. Idealization of a planar structure, such as a plate, reduces the number of peridynamic interactions to be solved. In this study, a peridynamic plate formulation of an orthotropic plate with transverse shear deformation is proposed. Peridynamic bond parameters are derived by localizing the peridynamic equations of motion and comparing them with equations of motion obtained using classical continuum mechanics. A failure theory is implemented in the formulation in order to perform crack propagation simulations. A MATLAB script is written using the proposed formulation for the solution of orthotropic plates under bending. The peridynamic solutions are obtained for both elastic deformations and failure prediction of pre-cracked plates, and are validated using finite-element analysis solutions.

Keywords

Peridynamics, Mindlin plate theory, composite plates, transverse shear deformation, bending, thick plates, orthotropy

1. Introduction

The latest aerospace designs commonly use composite materials in critical load-carrying parts because of their excellent strength-to-weight ratio [1]. Increasing the utilization of composite materials in structural designs requires the development of numerical tools to analyze their mechanical behavior [2–4]. In typical applications, carbon or glass fabrics are combined with relatively thick core material structures, such as honeycomb, to take advantage of their relatively high out-of-plane shear strength and stiffness [5].

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Classical laminate theory, a commonly used analysis tool for composite materials, is based on Kirchhoff's plate hypothesis. In Kirchhoff's plate theory, it is assumed that the normals to the mid-surface remain normal after deformation, which means that the transverse shear deformation is neglected [6]. Therefore, it is only applicable to thin plates. Because of the relatively high thickness of core materials, the transverse shear deformation of sandwich panels should not be neglected in the structural analysis of such aircraft components. Early studies concerning the transverse shear deformation were conducted by Reissner [7] and Mindlin [8]. Their approach was analogous to that of Timoshenko [9, 10], which was based on the assumption that planar sections remain planar after deformation. Mindlin–Reissner plate theory is based on the assumption of a linear strain function across the thickness, and is also known as first-order shear deformation theory. Whitney presented an approximate numerical solution for the laminated plates using this theory [11], and the theory was further extended to higher-order shear deformation theories using the polynomial strain function [12–15]. Mantari et al. [16] proposed a higher-order shear deformation theory, which can calculate the strain distributions accurately through the thickness without using a shear correction factor.

The aforementioned plate theories are commonly used in classical continuum mechanics (CCM) theory. This theory has shortcomings when it comes to model discontinuities or cracks in a structure. The governing equation of CCM theory requires spatial derivatives of the stress tensor that are not defined at the discontinuities in a structure. To overcome this restriction, Silling [17] proposed a nonlocal theory of continuum mechanics, called peridynamic theory. In the formulation of peridynamic theory, integro-differential equations are employed; this means that the equations are also valid at the discontinuities. Thus, a special treatment at the crack tip is not needed and the failure behavior of discontinuities is naturally captured [18]. Peridynamic formulations can be divided into two subcategories, namely a bond-based formulation and a state-based formulation [19]. The bond-based formulation assumes pairwise interaction between the material points, whereas the state-based formulation also accounts for indirect interactions between the material points [20]. In the bond-based formulation, the Poisson ratio of the material is limited to $1/3$ in the plane stress problems and $1/4$ in the plane strain problems because of the pairwise interactions. Despite this restriction, the bond-based formulation has still been employed in recent peridynamic studies [21, 22]. Zhu and Ni [21] reformulated the bond-based peridynamic theory with bond rotation enrichment to overcome the limitations of a fixed Poisson ratio. In their formulation, each bond has a normal stiffness and a tangential stiffness.

Peridynamic theory is a nonlocal theory with the advantage of an integral formulation that is also valid at defects such as cracks and voids. Therefore, it has been successfully applied to the failure analysis of both isotropic and anisotropic materials. Although crack branching in brittle materials is very unstable, it is well captured using peridynamic theory [23]. The simplest peridynamic constitutive model (linear microelasticity with brittle damage) is proven to be adequate for capturing the complex fracture patterns of a glass plate caused by the impact of a projectile [24]. Peridynamic theory is also successfully applied to nanofiber networks and membranes where transition from mode III fracture to mode I fracture is observed [25]. Ductile behavior of materials can also be modeled using peridynamic theory. Madenci and Oterkus [26] proposed a peridynamic constitutive relation for plastic deformation based on von Mises yield criteria with isotropic hardening. A Johnson–Cook material model that accounts for the rate-dependent plasticity with temperature effects is also implemented in peridynamic formulation, as presented by Amani et al. [27]. Although the peridynamic theory was originally introduced in 2000 by Silling [17], there has been an extensive peridynamic literature in different fields. A thorough review of peridynamic literature is presented by Javili et al. [28].

Significant efforts have been made to model structural behavior in composite materials using CCM theory and finite-element analysis (FEA) [29–31]. Those efforts addressed quasistatic loading conditions in common. Peridynamic theory is also a robust tool for studying failure propagation in composite materials under both static and dynamic conditions [32, 33]. Peridynamic material constants and failure parameters can be obtained using both elastic modulus and fracture toughness, as in Oterkus and Madenci [18]. Hu et al. [33] have developed a homogenized peridynamic model for unidirectional composites and have simulated dynamic fractures for different fiber orientations. Ghajari et al. [34] proposed a new peridynamic material model that enabled a continuous relationship between the fiber orientation and the bond properties such as the stiffness and the critical stretch. This feature enabled peridynamic simulations of anisotropic materials with random fiber orientations.

Peridynamic theory can also be implemented using commercial FEA codes [35–37]. The advantage of such an implementation is to couple the peridynamic and the FEA formulations, enabling us to analyze a complex structure. For instance, damage propagation in a stiffened curved composite panel, as in Oterkus et al. [38], and thermal shocks on ceramic materials, as in Giannakeas et al. [39], can be simulated by using the combined FEA and peridynamic approach. Other numerical methods, such as the finite-point method, can also be coupled with peridynamic theory for the solution of dynamic fracture problems, as in Shojaei et al. [40].

Numerical simulations of three-dimensional structures using nonlocal theories can be computationally costly. Thus, these structures can be idealized using two-dimensional models under certain assumptions. Such idealizations can also be applied to peridynamic simulations. Peridynamic equations of motions and peridynamic material parameters should also adapt to those idealizations. There are few studies demonstrating the advantages of two-dimensional peridynamic modeling in the literature. O'Grady and Foster [41] built a nonordinary state-based peridynamic model for the bending of flat Kirchhoff–Love plates. Taylor and Steigmann [42] proposed a two-dimensional peridynamic plate theory that could model failure due to bending. Taştan et al. [43] developed a peridynamic model based on Kirchhoff plate theory to be used in the failure modeling of orthotropic thin plates under bending. These plate formulations can only be applied to flat shells. Chowdhury et al. [44] introduced the curved bond concept and proposed a generalized two-dimensional surface-based peridynamic model that is also applicable to general curved shells. The symmetric boundary modeling approach in combination with the FEA formulation also reduces the execution cost of peridynamic solutions, as presented by Yaghoobi and Chorzepa [45].

A Mindlin plate peridynamic formulation capable of considering transverse shear deformations has been introduced in a recent paper by Diyaroglu et al. [46]. The mentioned study is closely related to the current study, in that the thick plate is modeled with isotropic material properties. To the best of the authors' knowledge, studies on the peridynamic formulation of thick plates having orthotropic material properties, taking into account transverse shear deformations, have not been reported in the open literature. In this study, we introduce a peridynamic orthotropic material formulation for thick plates using a homogenized approach. The new model will consider the bending stiffness in the two-dimensional formulation, extending the approach described in Diyaroglu et al. [46] to orthotropic materials. Orthotropic plate kinematics are introduced in Section 2.

2. Orthotropic plate kinematics

The geometry of bending a Mindlin plate is depicted in Figure 1. The rotations with respect to the x and the y axes at any point on the plate are denoted $\phi_{x(k)}$, $\phi_{y(k)}$, which can be defined in the following form:

$$\phi_{x(k)} = \frac{\partial w_{(k)}}{\partial x_{(k)}}, \quad (1)$$

$$\phi_{y(k)} = \frac{\partial w_{(k)}}{\partial y_{(k)}}, \quad (2)$$

where $w_{(k)}$ is the out-of plane displacement of a material point (k). The rotation of any material point (k), $\phi_{(k)}$, in an arbitrary axis $\mathbf{n}$ can then be written as

$$\phi_{(k)} = \phi_{x(k)} \cos \theta_{(k)} + \phi_{y(k)} \sin \theta_{(k)}. \quad (3)$$

Curvatures due to bending of the plate can be obtained by taking the derivative of the rotation with respect to the corresponding coordinate axis, as

$$\kappa_{x(k)} = \frac{1}{r_{x(k)}} = -\frac{\partial \phi_{x(k)}}{\partial x_{(k)}} = -\frac{\partial^2 w_{(k)}}{\partial x_{(k)}^2}, \quad (4)$$

$$\kappa_{y(k)} = \frac{1}{r_{y(k)}} = -\frac{\partial \phi_{y(k)}}{\partial y_{(k)}} = -\frac{\partial^2 w_{(k)}}{\partial y_{(k)}^2}, \quad (5)$$

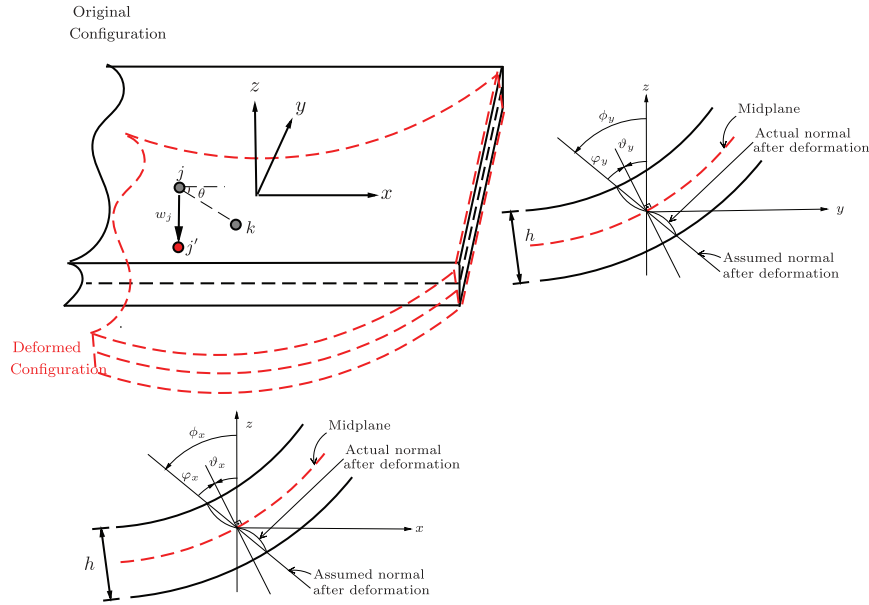


Figure 1. Original and deformed configurations of a Mindlin plate.

$$\kappa_{xy(k)} = \frac{1}{r_{xy(k)}} = -\frac{\partial^2 w_{(k)}}{\partial x_{(k)} \partial y_{(k)}}, \quad (6)$$

where $r_{x(k)}$, $r_{y(k)}$, and $r_{xy(k)}$ are the radii of curvature of material point (k) in the x and y directions and the twist radius of curvature, respectively. The curvature in an arbitrary direction can be defined as equations (4) to (6)

$$\kappa_{(k)} = \frac{1}{r_n} = -\frac{\partial}{\partial n} \left(\frac{\partial w_{(k)}}{\partial n} \right) = \kappa_{x(k)} \cos^2 \theta + 2\kappa_{xy(k)} \sin \theta \cos \theta + \kappa_{y(k)} \sin^2 \theta. \quad (7)$$

The curvature between two material points (j) and (k) can be expressed as

$$\kappa_{(k)(j)} = -\frac{\partial}{\partial n} \left(\frac{\partial w_{(k)(j)}}{\partial n} \right) = \kappa_{x(k)(j)} \cos^2 \theta + 2\kappa_{xy(k)(j)} \sin \theta \cos \theta + \kappa_{y(k)(j)} \sin^2 \theta. \quad (8)$$

It is assumed that the twisting component of the curvature field ($\kappa_{xy(k)}$) is zero under bending. Using this assumption, equation (8) becomes

$$\kappa_{(k)(j)} = \kappa_{x(k)(j)} \cos^2 \theta + \kappa_{y(k)(j)} \sin^2 \theta. \quad (9)$$

The partial derivatives in the curvature expressions given in equations (4) and (5) are discretized for the peridynamic formulation. The curvatures between material points at $x_{(j)}$ and $x_{(k)}$ can then be written as

$$\kappa_{x(k)(j)} = \frac{\phi_{x(j)} - \phi_{x(k)}}{x_{(j)} - x_{(k)}}, \quad (10)$$

$$\kappa_{y(k)(j)} = \frac{\phi_{y(j)} - \phi_{y(k)}}{y_{(j)} - y_{(k)}}, \quad (11)$$

where x_j , x_k , y_j , and y_k are the x and the y coordinates of material points j and k , respectively. Substituting equations (10) and (11) into equation (9), the curvature of material point (k) can be written as

$$\kappa_{(k)(j)} = \left(\frac{\phi_{x(j)} - \phi_{x(k)}}{\xi_x} \right) \cos^2 \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{\xi_y} \right) \sin^2 \theta, \quad (12)$$

where

$$\xi_x = x_{(j)} - x_{(k)} = \xi_{(k)(j)} \cos \theta, \quad (13)$$

$$\xi_y = y_{(j)} - y_{(k)} = \xi_{(k)(j)} \sin \theta. \quad (14)$$

In equations (13) and (14), $\xi_{(k)(j)}$ denotes the initial distance between material points (j) and (k) . Substituting equations (13) and (14) into equation (12) results in

$$\kappa_{(k)(j)} = \left(\frac{\phi_{x(j)} - \phi_{x(k)}}{\xi_{(k)(j)}} \right) \cos \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{\xi_{(k)(j)}} \right) \sin \theta. \quad (15)$$

As depicted in Figure 1, the transverse shear angles of material points (j) and (k) can be written as

$$\varphi_{(j)} = \vartheta_{(k)(j)} - \phi_{(j)}, \quad (16)$$

$$\varphi_{(k)} = \vartheta_{(k)(j)} - \phi_{(k)}, \quad (17)$$

where φ , ϑ , and ϕ denote the transverse shear angle, the slope between the material points and the rotations of the material points respectively. The slope between material points (j) and (k) can be written as

$$\vartheta_{(k)(j)} = \frac{w_{(j)} - w_{(k)}}{\xi_{(k)(j)}}. \quad (18)$$

The transverse shear angle between material points (j) and (k) can be written by averaging the transverse shear angles at material points (j) and (k) , as

$$\varphi_{(k)(j)} = \frac{w_{(j)} - w_{(k)}}{\xi_{(k)(j)}} - \frac{(\phi_{x(j)} + \phi_{x(k)}) \cos \theta + (\phi_{y(j)} + \phi_{y(k)}) \sin \theta}{2}. \quad (19)$$

3. Peridynamic theory

In CCM theory, the deformation of a body subjected to external loads can be calculated by treating the body as a continuum. Therefore, it is assumed that a continuous medium has an infinite number of infinitesimal volumes that interact only with their immediate neighbors. The equation of motion in CCM theory can be written as

$$\rho(\mathbf{x})\ddot{\mathbf{u}} + \nabla \sigma + \mathbf{b}(\mathbf{x}, t) = 0, \quad (20)$$

where $\rho(\mathbf{x})$, $\ddot{\mathbf{u}}$, and $\mathbf{b}(\mathbf{x}, t)$ denote the mass density, the spatial acceleration, and the body force density vector of the material point $\mathbf{x}$, respectively. The spatial derivatives of stress tensor σ required in equation (20) are not valid at the discontinuities of a structure, such as cracks and voids. In the peridynamic formulation, the equation of motion does not contain spatial derivatives and can be written as

$$\rho(\mathbf{x})\ddot{\mathbf{u}}(\mathbf{x}, t) = \int_H \mathbf{f}(\mathbf{u}' - \mathbf{u}, \mathbf{x}' - \mathbf{x}) dH + \mathbf{b}(\mathbf{x}, t), \quad (21)$$

where H is the horizon of the material point with a radius of δ , ρ is the mass density, and $\mathbf{u}$ and $\mathbf{b}$ denote the displacement and the body force vector fields, respectively. Note that $\mathbf{f}(\mathbf{u}(\mathbf{x}') - \mathbf{u}(\mathbf{x}), \mathbf{x}' - \mathbf{x})$ is the pairwise force vector that the material point $\mathbf{x}$ exerts on the material point $\mathbf{x}'$, as illustrated in Figure 2. It has been shown that the peridynamic theory converges to the classical theory of elasticity as the length of the horizon tends to zero [17].

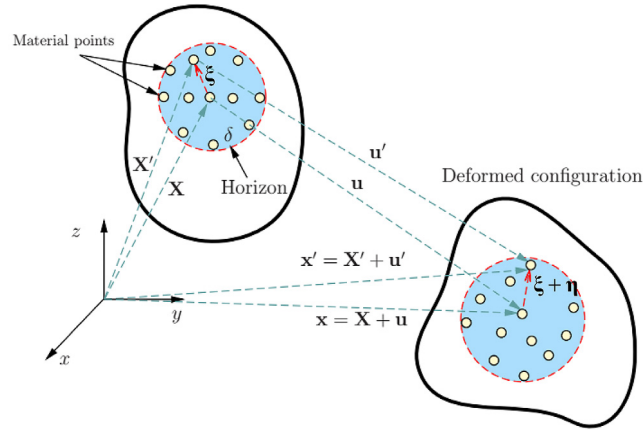


Figure 2. Geometry of the material points at the deformed and undeformed configurations.

For microelastic materials, the relation between the force density vector $\mathbf{f}$ and the scalar-valued function, ω , is defined by Silling [17] as

$$\mathbf{f}(\boldsymbol{\eta}, \boldsymbol{\xi}) = \frac{\partial}{\partial \boldsymbol{\eta}} \omega(\boldsymbol{\eta}, \boldsymbol{\xi}), \quad (22)$$

where ω is a pairwise potential function and is often termed the micropotential in the literature on peridynamic theory. In equation (22), the relative displacement vector, $\boldsymbol{\eta}$, and the relative position vector, $\boldsymbol{\xi}$, may be defined as

$$\boldsymbol{\eta} = \mathbf{u}' - \mathbf{u}, \quad (23)$$

$$\boldsymbol{\xi} = \mathbf{x}' - \mathbf{x}. \quad (24)$$

4. Equations of motion for an orthotropic Mindlin plate

In this section, the equations of motion for an orthotropic Mindlin plate will be given for both CCM and peridynamic formulations.

4.1. Classical continuum mechanics formulation

In CCM theory, the generalized form of the equations of motion for a Mindlin plate using the deep shell approach are given by Soedel [47] as follows:

$$\frac{\partial M_{xx}}{\partial x} + \frac{\partial M_{yx}}{\partial y} - Ghk^2 \epsilon_{xz} - \rho \frac{h^3}{12} \ddot{\phi}_x = 0, \quad (25)$$

$$\frac{\partial M_{xy}}{\partial x} + \frac{\partial M_{yy}}{\partial y} - Ghk^2 \epsilon_{yz} - \rho \frac{h^3}{12} \ddot{\phi}_y = 0, \quad (26)$$

$$-k^2 Gh \frac{\partial \epsilon_{xz}}{\partial x} - k^2 Gh \frac{\partial \epsilon_{yz}}{\partial y} + \rho h \ddot{w} = q_z, \quad (27)$$

where M_{ij} , ϕ_{ij} ($i, j = x, y$) denote the applied moment components at the plate, and x and y represent the in-plane coordinates. Moreover, k^2 , h , G , and q_z denote the shear correction factor, the plate thickness, the shear modulus, and the applied load in the z direction, respectively. The strain–displacement relations in the transverse direction can be written as

$$\epsilon_{xz} = \phi_x + \frac{\partial w}{\partial x}, \quad (28)$$

$$\epsilon_{yz} = \phi_y + \frac{\partial w}{\partial y}. \quad (29)$$

The moment–curvature relations of an orthotropic Mindlin plate can be written as

$$M_{xx} = D_{11}\kappa_{xx} + D_{12}\kappa_{yy}, \quad (30)$$

$$M_{yy} = D_{12}\kappa_{xx} + D_{22}\kappa_{yy}, \quad (31)$$

$$M_{xy} = D_{33}\kappa_{xy}, \quad (32)$$

where D_{ij} are the components of the bending stiffness matrix of the plate and the curvatures κ_{xx} , κ_{yy} , and κ_{xy} are defined in terms of the rotations as

$$\kappa_{xx} = \phi_{x,x}, \quad (33)$$

$$\kappa_{yy} = \phi_{y,y}, \quad (34)$$

$$\kappa_{xy} = \phi_{x,y} + \phi_{y,x}. \quad (35)$$

Therefore, the equations of motion for an orthotropic Mindlin plate in CCM theory can be obtained by using equations (30) to (35) in equations (25) to (27), as

$$\rho \frac{h^3}{12} \ddot{\phi}_x = D_{11}\phi_{x,xx} + (D_{12} + D_{33})\phi_{y,xy} + D_{33}\phi_{x,yy} - Ghk^2(\phi_x - w_{,x}), \quad (36)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_y = D_{22}\phi_{y,yy} + (D_{12} + D_{33})\phi_{x,xy} + D_{33}\phi_{y,xx} - Ghk^2(\phi_y - w_{,y}), \quad (37)$$

$$\rho h \ddot{w}_{(k)} - hk^2 G(w_{,xx} + w_{,yy} - \phi_{x,x} - \phi_{y,y}) = q_z, \quad (38)$$

where

$$D_{11} = \frac{E_{11}}{12(1 - \nu_{12}\nu_{21})}, \quad (39)$$

$$D_{22} = \frac{E_{22}}{12(1 - \nu_{12}\nu_{21})}, \quad (40)$$

$$D_{12} = \frac{\nu_{21}E_{11}}{12(1 - \nu_{12}\nu_{21})}, \quad (41)$$

$$D_{33} = \frac{G_{12}}{12}. \quad (42)$$

4.2. Peridynamic formulation

4.2.1. Equation of motion using peridynamic theory. The principle of virtual work is applied to obtain the peridynamic equations of motion for an orthotropic plate subjected to bending. The total kinetic energy of the orthotropic plate due to bending in terms of the displacements of a material point k , $(u_{(k)}, v_{(k)}, w_{(k)})$ can be written as

$$T = \frac{1}{2} \rho \sum_{k=1}^{\infty} [\dot{u}_{(k)}^2 + \dot{v}_{(k)}^2 + \dot{w}_{(k)}^2] V_{(k)}, \quad (43)$$

where $V_{(k)}$ and ρ represent the infinitesimally small incremental volume of the reference material point (k) and the mass density of the material, respectively. The in-plane displacement components of the reference material point $u_{(k)}$ and $v_{(k)}$ can now be written in terms of rotations as

$$u_{(k)} = -z\phi_{x(k)}, \quad (44)$$

$$v_{(k)} = -z\phi_{y(k)}. \quad (45)$$

Substituting equations (44) and (45) into equation (43), the total kinetic energy can be obtained as

$$T = \frac{1}{2}h\rho \sum_{k=1}^{\infty} \left(\dot{w}_{(k)}^2 + \frac{h^2}{12} \dot{\phi}_{x(k)}^2 + \frac{h^2}{12} \dot{\phi}_{y(k)}^2 \right) A_{(k)}, \quad (46)$$

where $A_{(k)}$ represents the area of the reference material point (k) and h denotes the thickness of the plate.

In this study, pure bending and transverse shear deformations are considered. Therefore, two different micropotential functions are defined, $\tilde{w}(\kappa)$ and $\hat{w}(\varphi)$. The relationship between the moment density function $\tilde{f}(\kappa)$ and the micropotential function $\tilde{w}(\kappa)$ and the relationship between the transverse force density function $\hat{f}(\varphi)$ and the micropotential function $\hat{w}(\varphi)$ can be written as

$$\tilde{f}_{(k)(j)}(\kappa_{(k)(j)}) = \frac{1}{\xi_{(j)(k)}} \frac{\partial \tilde{w}_{(k)(j)}(\kappa_{(k)(j)})}{\partial (\kappa_{(k)(j)})}, \quad (47)$$

$$\hat{f}_{(k)(j)}(\varphi_{(k)(j)}) = \frac{1}{\xi_{(j)(k)}} \frac{\partial \hat{w}_{(k)(j)}(\varphi_{(k)(j)})}{\partial (\varphi_{(k)(j)})}. \quad (48)$$

For linear elastic materials, the peridynamic moment density and the transverse force density due to bending can be written as

$$\tilde{f}_{(j)(k)}(\kappa) = c_b \kappa_{(j)(k)}, \quad (49)$$

$$\hat{f}_{(j)(k)}(\varphi) = c_s \varphi_{(j)(k)}, \quad (50)$$

where c_b and c_s are peridynamic material constants.

Substituting equations (49) and (50) into equations (47) and (48), and integrating with respect to the curvature and the transverse shear field, the micropotentials may be obtained as

$$\tilde{w}_{(k)(j)}(\kappa_{(k)(j)}) = \frac{1}{2} c_b \kappa_{(k)(j)}^2 \xi_{(j)(k)}, \quad (51)$$

$$\hat{w}_{(k)(j)}(\varphi_{(k)(j)}) = \frac{1}{2} c_s \varphi_{(k)(j)}^2 \xi_{(j)(k)}. \quad (52)$$

The total potential energy can then be obtained by summation of the micropotentials, as

$$\begin{aligned} U = & \sum_{i=1}^{\infty} \left\{ \frac{1}{2} \sum_{j=1}^{\infty} \frac{1}{2} [\tilde{w}_{(i)(j)}(\kappa_{(i)(j)}) + \tilde{w}_{(j)(i)}(\kappa_{(j)(i)})] V_{(j)} - \tilde{b}_{(i)} \phi_{(i)} \right\} V_{(i)} \\ & + \sum_{i=1}^{\infty} \left\{ \frac{1}{2} \sum_{j=1}^{\infty} \frac{1}{2} [\hat{w}_{(i)(j)}(\varphi_{(i)(j)}) + \hat{w}_{(j)(i)}(\varphi_{(j)(i)})] V_{(j)} - \hat{b}_{(i)} w_{(i)} \right\} V_{(i)}, \end{aligned} \quad (53)$$

where $\hat{b}_i$ and $\tilde{b}_i$ denote the body forces and the body moments, respectively. The independent variables in the kinetic energy and the total potential energy of the system are $\dot{w}_k$, w_k , $\phi_{\alpha k}$, and $\phi_{\alpha k}$. Thus, Euler–Lagrange equations can be written as follows:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{w}_{(k)}} - \frac{\partial L}{\partial w_{(k)}} = 0, \quad (54)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\phi}_{\alpha(k)}} - \frac{\partial L}{\partial \phi_{\alpha(k)}} = 0, \quad (55)$$

where $\alpha = x, y$. Differentiating using the Lagrangian definition ($L = T - U$), the following equations of motion are obtained:

$$\rho h \ddot{w}_{(k)} + \sum_{j=1}^{\infty} \frac{1}{2} \left[\xi_{jk} \hat{f}_{(k)(j)} \frac{\partial \varphi_{(k)(j)}}{\partial (w_{(k)})} + \xi_{jk} \hat{f}_{(j)(k)} \frac{\partial \varphi_{(j)(k)}}{\partial (w_{(k)})} \right] V_{(j)} - \hat{b}_{\alpha(k)} = 0, \quad (56)$$

$$\begin{aligned} \rho \frac{h^3}{12} \ddot{\phi}_{\alpha(k)} + \sum_{j=1}^{\infty} \frac{1}{2} \xi_{jk} \left[\tilde{f}_{(k)(j)} \frac{\partial (\kappa_{(k)(j)})}{\partial (\phi_{\alpha(k)})} + \tilde{f}_{(j)(k)} \frac{\partial (\kappa_{(j)(k)})}{\partial (\phi_{\alpha(k)})} \right] V_{(j)} \\ + \sum_{j=1}^{\infty} \frac{1}{2} \xi_{jk} \left[\hat{f}_{(k)(j)} \frac{\partial \varphi_{(k)(j)}}{\partial (\phi_{\alpha(k)})} + \hat{f}_{(j)(k)} \frac{\partial \varphi_{(j)(k)}}{\partial (\phi_{\alpha(k)})} \right] V_{(j)} - \tilde{b}_{\alpha(k)} = 0. \end{aligned} \quad (57)$$

By taking the derivatives in equations (56) and (57) and using the force density functions in equations (49) and (50), equations (56) and (57) become

$$\rho h \ddot{w}_{(k)} = c_s \sum_{j=1}^{\infty} \varphi_{(k)(j)} V_{(j)}, \quad (58)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{x(k)} = c_b \sum_{j=1}^{\infty} \kappa_{(k)(j)} \cos \theta V_{(j)} + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \varphi_{(k)(j)} \cos \theta V_{(j)}, \quad (59)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{y(k)} = c_b \sum_{j=1}^{\infty} \kappa_{(k)(j)} \sin \theta V_{(j)} + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \varphi_{(k)(j)} \sin \theta V_{(j)}. \quad (60)$$

Substitution of equations (15) and (19) into equations (58) to (60) yields the following equations of motion:

$$\rho h \ddot{w}_{(k)} = c_s \sum_{j=1}^{\infty} \left(\frac{w_{(j)} - w_{(k)}}{\xi_{(j)(k)}} - \frac{\phi_{x(j)} + \phi_{x(k)}}{2} \cos \theta - \frac{\phi_{y(j)} + \phi_{y(k)}}{2} \sin \theta \right) V_{(j)}, \quad (61)$$

$$\begin{aligned} \rho \frac{h^3}{12} \ddot{\phi}_{x(k)} = c_b \sum_{j=1}^{\infty} \left[\left(\frac{\phi_{x(j)} - \phi_{x(k)}}{\xi_{jk}} \right) \cos \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{\xi_{jk}} \right) \sin \theta \right] \cos \theta V_{(j)} \\ + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \left(\frac{w_j - w_k}{\xi_{jk}} - \frac{\phi_{x(j)} + \phi_{x(k)}}{2} \cos \theta - \frac{\phi_{y(j)} + \phi_{y(k)}}{2} \sin \theta \right) \cos \theta V_{(j)}, \end{aligned} \quad (62)$$

$$\begin{aligned} \rho \frac{h^3}{12} \ddot{\phi}_{y(k)} = c_b \sum_{j=1}^{\infty} \left[\left(\frac{\phi_{x(j)} - \phi_{x(k)}}{\xi_{jk}} \right) \cos \theta + \left(\frac{\phi_{y(j)} - \phi_{y(k)}}{\xi_{jk}} \right) \sin \theta \right] \sin \theta V_{(j)} \\ + \frac{1}{2} c_s \sum_{j=1}^{\infty} \xi_{jk} \left(\frac{w_j - w_k}{\xi_{jk}} - \frac{\phi_{x(j)} + \phi_{x(k)}}{2} \cos \theta - \frac{\phi_{y(j)} + \phi_{y(k)}}{2} \sin \theta \right) \sin \theta V_{(j)}. \end{aligned} \quad (63)$$

4.2.2. Determination of peridynamic material parameters. For composite materials, the peridynamic material parameter $c_b(\theta)$ is an elastic constant depending on material orientation. It can be written as

$$c_b(\theta) = \begin{cases} c_{bf} & \text{for } \theta = \theta_f \\ c_{bm} & \text{for } \theta \neq \theta_f \end{cases}, \quad (64)$$

where θ_f is the fiber orientation angle and c_{bf} and c_{bm} denote the peridynamic material parameters associated with bending deformation for the fiber and the matrix bonds, respectively. It should be noted that there exists a linear relationship between the peridynamic moment density and the curvature for both fiber and matrix bonds, independently of each other. To obtain peridynamic material parameters, the equations of motion given in equations (61) to (63) should approximate to their classical counterparts. Thus, the deflections and the rotations can be written using the Taylor series expansion, as

$$\begin{aligned} w_{(j)} = & w_{(k)} + w_{(k),x} \xi_{(j)(k)} \cos \theta + w_{(k),y} \xi_{(j)(k)} \sin \theta + \frac{1}{2} w_{(k),xx} \xi_{(j)(k)}^2 \cos^2 \theta \\ & + \frac{1}{2} w_{(k),yy} \xi_{(j)(k)}^2 \sin^2 \theta + w_{(k),xy} \xi_{(j)(k)}^2 \sin \theta \cos \theta, \end{aligned} \quad (65)$$

$$\begin{aligned} \phi_{x(j)} = & \phi_{x(k)} + \phi_{x(k),x} \xi_{(j)(k)} \cos \theta + \phi_{x(k),y} \xi_{(j)(k)} \sin \theta + \frac{1}{2} \phi_{x(k),xx} \xi_{(j)(k)}^2 \cos^2 \theta \\ & + \frac{1}{2} \phi_{x(k),yy} \xi_{(j)(k)}^2 \sin^2 \theta + \phi_{x(k),xy} \xi_{(j)(k)}^2 \sin \theta \cos \theta, \end{aligned} \quad (66)$$

$$\begin{aligned} \phi_{y(j)} = & \phi_{y(k)} + \phi_{y(k),x} \xi_{(j)(k)} \cos \theta + \phi_{y(k),y} \xi_{(j)(k)} \sin \theta + \frac{1}{2} \phi_{y(k),xx} \xi_{(j)(k)}^2 \cos^2 \theta \\ & + \frac{1}{2} \phi_{y(k),yy} \xi_{(j)(k)}^2 \sin^2 \theta + \phi_{y(k),xy} \xi_{(j)(k)}^2 \sin \theta \cos \theta. \end{aligned} \quad (67)$$

By using equations (65) to (67) in equations (61) to (63), the peridynamic equations of motion can be written as

$$\rho h \ddot{w}_{(k)} = \frac{\pi}{2} c_s h \frac{\delta^3}{3} (w_{(k),xx} + w_{(k),yy} - \phi_{x(k),x} - \phi_{y(k),y}), \quad (68)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{x(k)} = \frac{\pi}{16} h \delta^3 \left(c_{bf} \frac{2}{3} - c_s \frac{\delta^2}{10} \right) [3\phi_{x(k),xx} + \phi_{x(k),yy} + 2\phi_{y(k),xy}] - \frac{\pi}{2} c_s h \frac{\delta^3}{3} (-w_{(k),x} + \phi_{x(k)}), \quad (69)$$

$$\rho \frac{h^3}{12} \ddot{\phi}_{y(k)} = \frac{\pi}{16} h \delta^3 \left(c_{bm} \frac{2}{3} - c_s \frac{\delta^2}{10} \right) [3\phi_{y(k),yy} + \phi_{y(k),xx} + 2\phi_{x(k),yx}] - \frac{\pi}{2} c_s h \frac{\delta^3}{3} (-w_{(k),y} + \phi_{y(k)}). \quad (70)$$

Note that the equations of motion in CCM theory for orthotropic plates with transverse shear deformations have been given previously, in equations (36) to (38). By comparing the terms in equations (36) to (38) and equations (68) to (70), the peridynamic material parameters can be determined as

$$c_s = \frac{6k^2 G}{\pi \delta^3}, \quad (71)$$

$$c_{bf} = \frac{1}{\pi \delta} \left(\frac{8D_{11}}{h \delta^2} + \frac{9kG}{10} \right), \quad (72)$$

$$c_{bm} = \frac{1}{\pi \delta} \left(\frac{8D_{22}}{h \delta^2} + \frac{9kG}{10} \right), \quad (73)$$

where k^2 is a shear correction factor and is taken as $\pi^2/12$ [46]. Equating the terms in equations (36) to (38) and equations (68) to (70) also indicates the constraint on the Poisson ratio as $\nu_{12} = \nu_{21} = 1/3$.

4.3. Orthotropic to isotropic conversion

Another check on the formulation can be made by converting the orthotropic equations to their isotropic counterparts. The peridynamic constants for isotropic materials can be obtained by taking $E_{11} = E_{22} = E$ and $D_{11} = D_{22} = D$ in equations (71) and (72) as follows:

$$c_s = \frac{6k^2 G}{\pi \delta^3}, \quad (74)$$

$$c_{bf} = c_{bm} = c_b = \frac{1}{\pi \delta} \left(\frac{8D}{h\delta^2} + \frac{9kG}{10} \right), \quad (75)$$

where

$$G = \frac{E}{2(1 + \nu)}, \quad (76)$$

$$D = \frac{Eh^3}{12(1 - \nu^2)}. \quad (77)$$

Substituting equations (76) and (77) into equations (74) and (75), the material constants for isotropic materials can be written as follows:

$$c_s = \frac{9Ek^2}{4\pi\delta^3}, \quad (78)$$

$$c_b = \frac{E}{\pi\delta} \left(\frac{3h_2}{4\delta^2} + \frac{27}{80}k^2 \right). \quad (79)$$

Equations (78) and (79) are the expressions for the peridynamic material constants of isotropic Mindlin plates, which were also derived by Diyaroglu et al. [46].

5. Failure implementation

In the peridynamic theory, failure can be implemented by irreversible breakage of the bonds between material points. In this study, the original approach given by Silling [17] is adopted. The curvatures and the transverse shear angles for an orthotropic plate subjected to bending need to be evaluated for the failure implementation at each time step. In the present work, bond breakage was related to the critical curvatures and the critical transverse shear angle, as reported in Diyaroglu et al. [46]. In that study, the critical curvature κ_c and critical transverse shear angle φ_c are obtained as

$$\kappa_c = \sqrt{\frac{4G_{IC}}{c_b h \delta^4}}, \quad (80)$$

$$\varphi_c = \sqrt{\frac{4G_{IIIC}}{c_s h \delta^4}}, \quad (81)$$

where G_{IC} and G_{IIIC} denote the Mod I and Mod III energy release rates, respectively. In this study, this formulation is extended to orthotropic plates by defining the critical curvatures in the fiber and matrix directions and the transverse shear angle of the orthotropic plates as follows:

$$\kappa_{c, \text{fiber}} = \sqrt{\frac{4G_1^{\text{fiber}}}{c_{bf} h \delta^4}}, \quad (82)$$

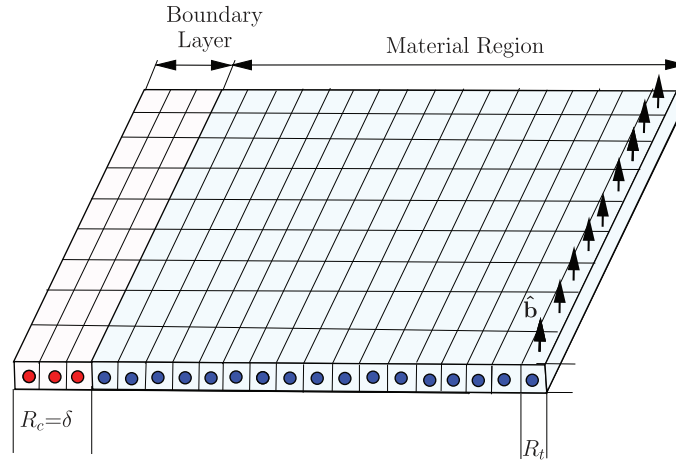


Figure 3. Application of boundary conditions and external loads on the peridynamic model.

$$\kappa_{c, \text{matrix}} = \sqrt{\frac{4G_{\text{I}}^{\text{matrix}}}{c_{\text{bm}}h\delta^4}}, \quad (83)$$

$$\varphi_c = \sqrt{\frac{4G_{\text{III}}}{c_s h \delta^4}}, \quad (84)$$

where $\kappa_{c, \text{fiber}}$ is the critical curvature for the bonds aligned in the fiber direction, while, for the other bonds, the critical curvature is $\kappa_{c, \text{matrix}}$. φ_c is the critical transverse shear angle. $G_{\text{I}}^{\text{fiber}}$, $G_{\text{I}}^{\text{matrix}}$, and G_{III} are the critical Mod I energy release rates of an orthotropic plate in the fiber and matrix directions and the critical Mod III energy release rate, respectively.

For crack growth analysis, the force density ($\hat{f}$) and moment density ($\tilde{f}$) functions are multiplied by a history-dependent damage function μ to realize the bond breakage. The damage function μ takes a value of either 1 or 0, as follows:

$$\mu(t_n, \theta) = \begin{cases} 1 & \text{for } (\kappa_{xx} \leq \kappa_{c, \text{fiber}}) \text{ and } (\kappa_{yy} \leq \kappa_{c, \text{matrix}}) \text{ and } (\varphi \leq \varphi_c) \\ 0 & \text{Otherwise} \end{cases}. \quad (85)$$

where t_n denotes the n th time step. The damage at a material point is then calculated as the ratio of the number of broken bonds to the total number of bonds in the horizon.

6. Application of the boundary conditions to the peridynamic model

The force loading conditions and the displacement boundary conditions can easily be applied in CCM theory by choosing the boundary and specifying the imposed conditions. However, in peridynamic theory, this is not the case. Although “natural boundary conditions” appear automatically in CCM theory, no natural boundary conditions arise in the derivation of peridynamic equilibrium equations [48]. Moreover, traction boundary conditions are not applied in the peridynamic approach. Instead, the transverse forces and the bending moments can be applied as body loads on the region R_t within the material shown in Figure 3. The displacement boundary conditions in peridynamic theory can be applied through a fictitious region. For example, constraints on the rotations ϕ_x , ϕ_y and out-of-plane displacement, w , along the boundary are applied on the fictitious region R_c illustrated in Figure 3. The size of the fictitious boundary layer, R_c , suggested by Silling and Askari [48] is taken as $\delta = 3\Delta x$ in this study.

7. Numerical solution

Dynamic relaxation methods are commonly used to solve nonlinear static problems. Dynamic relaxation methods provide iterative solutions that converge to a steady state representing the static condition. In dynamic relaxation methods, nonlinear equations are solved by introducing an artificial damping into the problem. The adaptive dynamic relaxation (ADR) method introduced by Underwood [49] is a type of dynamic relaxation solution method that changes the damping coefficient adaptively at each time step. Kilic and Madenci [50] associated the ADR method with peridynamic theory and used the ADR algorithm to solve the governing peridynamic equations of basic structural components under tensile, bending, and thermal static loading conditions. In the present study, a similar approach is adopted and the ADR solution scheme is used to solve the equations of motion given in equations (68) to (70). Artificial damping terms c_n and d_n are introduced in the equations of motion as

$$\ddot{\phi}_{(k)}(t_n) + c_n \phi_{(k)}(t_n) = \frac{12 \tilde{f}_{(k)}(t_n)}{h^3 \lambda_{(k)}}, \quad (86)$$

$$\ddot{w}_{(k)}(t_n) + d_n w_{(k)}(t_n) = \frac{\hat{f}_{(k)}(t_n)}{\lambda_{(k)}}, \quad (87)$$

where t_n and λ_k denote the n th time step and the modified density, respectively. The artificial damping coefficients c_n and d_n are calculated adaptively at each time step by considering the rotations and the out-of-plane displacements of all material points as

$$c_n = 2 \sqrt{\frac{(\underline{\phi}^n)^T (\underline{K}^n) (\underline{\phi}^n)}{(h^3/12) (\underline{\phi}^n)^T (\underline{\phi}^n)}}, \quad (88)$$

$$d_n = 2 \sqrt{\frac{(\underline{w}^n)^T (\underline{L}^n) (\underline{w}^n)}{(\underline{w}^n)^T (\underline{w}^n)}}, \quad (89)$$

where

$$(\underline{\phi}^n)^T = [\phi_{(1)}, \phi_{(2)}, \dots, \phi_{(n)}], \quad (90)$$

$$(\underline{w}^n)^T = [w_{(1)}, w_{(2)}, \dots, w_{(n)}]. \quad (91)$$

and $\underline{K}^n$ and $\underline{L}^n$ denote diagonal local stiffness matrices. They can be expressed as

$$K_{kk}^n = \left(\frac{\tilde{f}_{(k)}^n}{\lambda_k} - \frac{\tilde{f}_{(k)}^{n-1}}{\lambda_k} \right) / \dot{\phi}_{(k)}^{n-1/2}, \quad (92)$$

$$L_{kk}^n = \left(\frac{\hat{f}_{(k)}^n}{\lambda_k} - \frac{\hat{f}_{(k)}^{n-1}}{\lambda_k} \right) / \dot{w}_{(k)}^{n-1/2}. \quad (93)$$

The terms of the diagonal mass matrix $\lambda_{(ii)}$ should fulfill Gresghorin's theorem as

$$\lambda_{ii} \geq \frac{1}{4} \Delta t^2 \sum_j K_{ij}, \quad (94)$$

where K_{ij} are the components of the stiffness matrix of the constructed equation system. The time step size $\Delta t = 1$ is a convenient value for ADR solutions [50]. By applying the central difference explicit

integration scheme, the rotational velocities $\dot{\phi}_k$, the rotations ϕ_k , the out-of-plane velocity $\dot{w}_k$, and the displacement w can be calculated as follows:

$$\dot{\phi}_k^{n+1/2} = \left(2 - c_n \dot{\phi}_k^{n-1/2} + \frac{2(\tilde{f}_{(k)}^n + \tilde{b}_{(k)}^n)}{\lambda_k(h^3/12)} \right) / (2 + c_n), \quad (95)$$

$$\dot{w}_k^{n+1/2} = \left(2 - d_n \dot{w}_k^{n-1/2} + \frac{2(\hat{f}_{(k)}^n + \hat{b}_{(k)}^n)}{\lambda_k} \right) / (2 + d_n), \quad (96)$$

$$\phi_k^{n+1} = \phi_k^n + \dot{\phi}_k^{n+1/2} \Delta t, \quad (97)$$

$$w_k^{n+1} = w_k^n + \dot{w}_k^{n+1/2} \Delta t, \quad (98)$$

$$\phi_k^{1/2} = \frac{(\Delta t) \tilde{b}_{(k)}}{2\lambda_k}, \quad (99)$$

$$w_k^{1/2} = \frac{(\Delta t) \hat{b}_{(k)}}{2\lambda_k}. \quad (100)$$

The application of the ADR method to solve static problems requires a choice of termination criteria. As the static solution corresponds to the steady-state solution, it is necessary to ensure that the solution is convergent. In the current study, a criterion based on the maximum out-of-plane displacement is defined as

$$\frac{w_{\max}(i) - w_{\max}(i-1)}{w_{\max}(i)} = e, \quad (101)$$

where e is a pre-selected small number and is taken as 0.005 and $w_{\max}(i)$ is the maximum out-of-plane displacement at time step i .

The horizon size, δ , has a significant effect on the peridynamic solutions. In this study, it is taken as $\delta = 3.015\Delta x$, as suggested by Silling and Askari [48]. The discrepancy between the analytical results and the peridynamic calculations increases as the radius of the horizon increases, owing to excessive wave dispersions. Higher horizon sizes also increase the computational time. δ values less than $3\Delta x$ fail to capture crack branching patterns as the cracks propagate along the rows or the columns of the material points [48].

8. Results and discussion

In this section, we provide some benchmark cases to validate our peridynamic model for orthotropic plates. A three-dimensional FEA model is generated for the validation of proposed two-dimensional peridynamic plate model depicted in Figure 4. FEA solutions are performed using commercially available software ABAQUS by employing the C8D8R element type. The numerical results are obtained for an isotropic plate subjected to transverse loading, as also studied by Diyaroglu et al. [46]. Next, we study orthotropic plates under transverse loading and uniform pressure. Subsequently, crack propagation studies are performed for orthotropic plates with central cracks. Note that the strength of our study comes from the fact that three-dimensional orthotropic plates can be successfully studied using two-dimensional peridynamic plate formulation. Now we analyze several different problems using our peridynamic model.

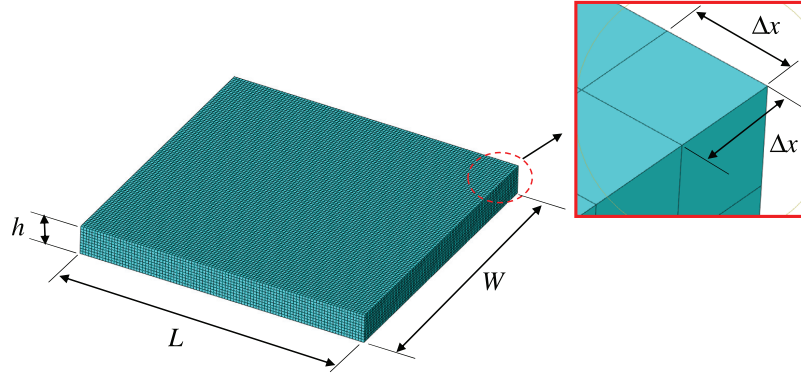


Figure 4. Finite-element analysis model of the Mindlin plate and its discretization.

8.1. Mindlin plate subjected to a transverse force

First, a square plate subjected to a transverse force is considered. The dimensions of the plate are $L = W = 1$ m and $h = 0.1$ m. The plate is fixed at the left end and the transverse load is introduced at the right end as illustrated in Figure 5. The transverse force is applied as body force of $\hat{\mathbf{b}} = 5 \times 10^8 \text{ N/m}^2 \mathbf{k}$ on a single row of material points at the right end. In both FEA and peridynamic calculations, the discretization size Δx is taken as 0.01 m. To demonstrate the convergence of the formulation to isotropic formulation, this problem is initially solved for an isotropic material. Then the same problem is solved for four different orthotropic plates.

8.1.1. Isotropic case. The proposed peridynamic formulation of the orthotropic Mindlin plate is transformed into an isotropic formulation by eliminating the directional dependency and taking $E_{11} = E_{22} = E$ and $D_{11} = D_{22} = D$. The elastic constants of the material are specified as $E = 200$ GPa and $\nu = 1/3$. The problem geometry and boundary conditions were given previously. The FEA and the peridynamic contours of the out-of-plane displacements w are shown in Figure 6. The rotations, ϕ , and out-of-plane displacements, w , along the $y = 0$ line on the plate are depicted in Figure 7, where the peridynamic results are compared with the peridynamic solutions given by Diyaroglu et al. [46] and the FEA solutions. The comparison indicates that the proposed orthotropic formulation converges to the isotropic formulation, as expected. The agreement between the FEA solution and Diyaroglu et al. [46] also indicate the validity of the FEA model, which is considered the baseline for further orthotropic cases.

8.1.2. Orthotropic case. In this part, the bending of orthotropic plates subjected to a transverse load is studied. The geometry, boundary conditions, and loading are the same as for the isotropic case. Three different orthotropic cases with $\theta_f = 0^\circ$ and one case for $\theta_f = 45^\circ$ are generated to demonstrate the capability of the proposed theory. The material properties and plate orientations are changed between the four cases, as defined in Table 1. The FEA model previously validated in the isotropic case is used to validate the orthotropic cases. The distributions of the out-of-plane displacement, w , of the plates obtained by the peridynamic and the FEA solutions for the first three cases are shown in Figure 8. The out-of-plane displacements, w , and the rotations, ϕ , along the $y = 0$ line on the plate obtained for the peridynamic and FEA solutions are also compared in Figure 9. It can be noted that the peridynamic and FEA solutions of with $\theta_f = 0^\circ$ oriented plates agree in the comparison of the results of the peridynamic model and FEA.

To elucidate the applicability of the proposed peridynamic model for a 45° plate, we modeled a 45° oriented plate under a transverse load. The material properties of the plate are given in Table 1. The loading, dimensions, and boundary conditions are the same as for the previous cases, as shown in Figure 5. The distribution of the out-of-plane displacement, w , obtained using the peridynamic and FEA solutions for $\theta_f = 45^\circ$ is depicted in Figure 10. The out-of-plane displacements, w , and rotations, ϕ , along the $y = 0$ line on the plate obtained using the peridynamic and FEA solutions are compared in Figure 11.

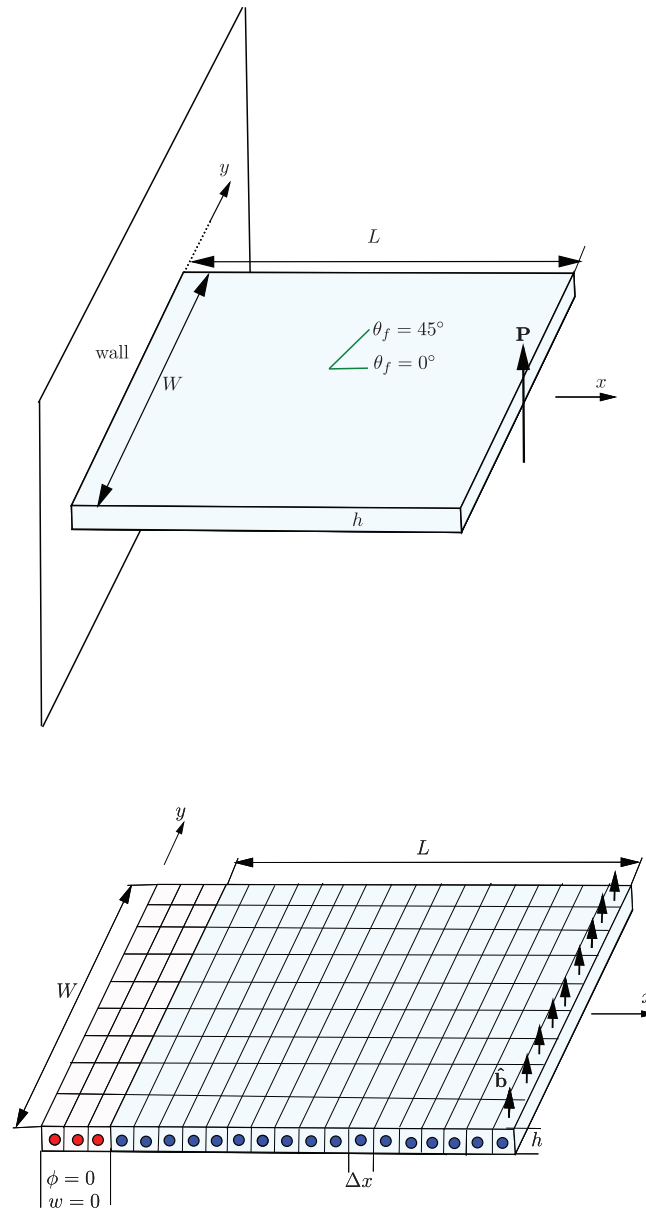


Figure 5. Mindlin plate subjected to transverse force.

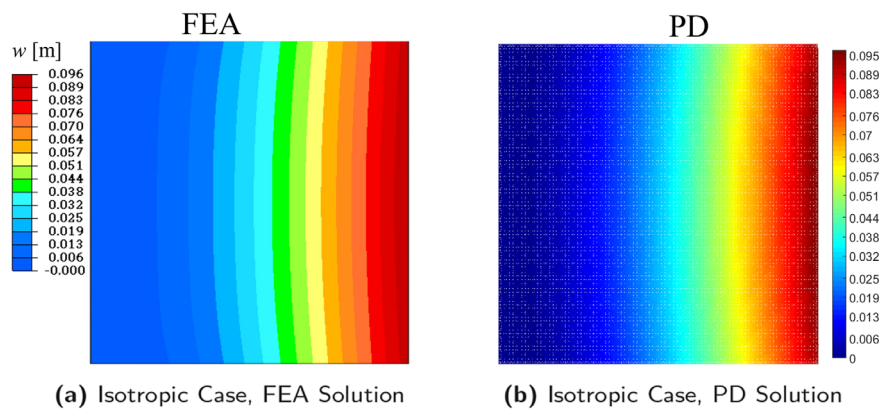


Figure 6. Comparison of out-of-plane displacement contours of an isotropic plate subjected to transverse loading: (a) finite-element analysis (FEA); (b) peridynamic (PD) theory.

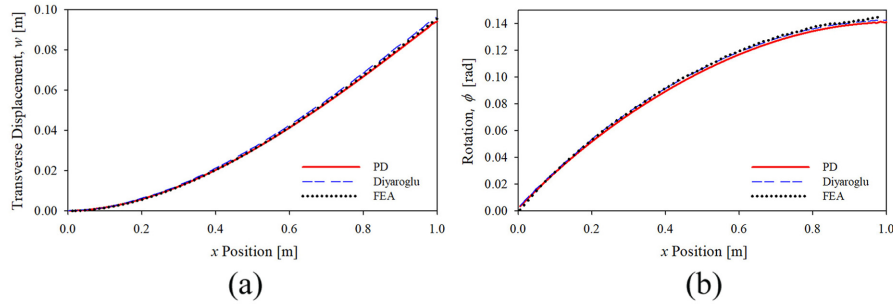


Figure 7. Displacements and rotations along the $y = 0$ line of an isotropic plate subjected to transverse loading. FEA: finite-element analysis; PD: peridynamic.

Table 1. Plate properties subjected to a transverse load.

Case	E_1 (GPa)	E_2 (GPa)	G (GPa)	θ_f (°)
1	200	20	75	0
2	180	10	67.5	0
3	90	50	33.75	0
4	220	150	82.5	45

We compared our two-dimensional peridynamic results with three-dimensional FEA solutions. It can be stated that the peridynamic solution captures the bending deformation of a 45° oriented plate fairly well. However, the discrepancy between the peridynamic and FEA results for Case 4 (45° oriented plate) is greater than that in Cases 1 to 3 (0° oriented plates). This is expected, because the bending deformation is coupled with a twisting deformation for an unbalanced composite plate. Note that the proposed formulation does not consider the bending–twisting coupling since it is assumed that the twisting component of the curvature field ($\kappa_{xy(k)}$) is zero under bending in equation (8).

8.2 Mindlin plate subjected to uniform pressure

In this part, the boundary conditions of the problem studied in Section 8.1 are changed to show the effect of different boundary conditions. A square plate fixed at both ends, as shown in Figure 12, and subjected to uniform pressure is considered. The rotations and translations are constrained at the fixed edges. The dimensions of the plate are $L = W = 1$ m and $h = 0.1$ m. In both FEA and peridynamic calculations, the discretization size Δx is taken as 0.01. Again, a fictitious region with a size of $3\Delta x$ is generated in the peridynamic solutions to apply the nodal constraints. The plate is subjected to a uniform body force $\mathbf{b} = 5 \times 10^8 \text{ N/m}^2 \mathbf{k}$, as shown in Figure 12. Two different cases are generated, as defined in Table 2, to check the dependency of the peridynamic formulation on material properties. The distributions of the out-of-plane displacement, w , obtained using peridynamic and FEA simulations for both cases are shown in Figure 13. The out-of-plane displacements, w , and the rotations, ϕ , along the $y = 0$ line on the plate obtained using peridynamic and FEA solutions are also compared in Figure 14. It can be seen that both peridynamic results are in good agreement with the FEA results.

8.3 Plate with pre-existing crack subjected to transverse load and uniform pressure

The propagation of pre-existing cracks within orthotropic plates is solved using the proposed formulation. The dimensions of the plate are $L = W = 1$ m and $h = 0.1$ m and the size of the central crack is $2a = 20$ mm. The material parameters are listed in Tables 3 and 4. First, the left side of the plate is constrained and the other side of the plate is subjected to a body force on one row of material points, as illustrated in Figure 15. The problem is solved for orthotropic plates oriented at 0° and 90° that are

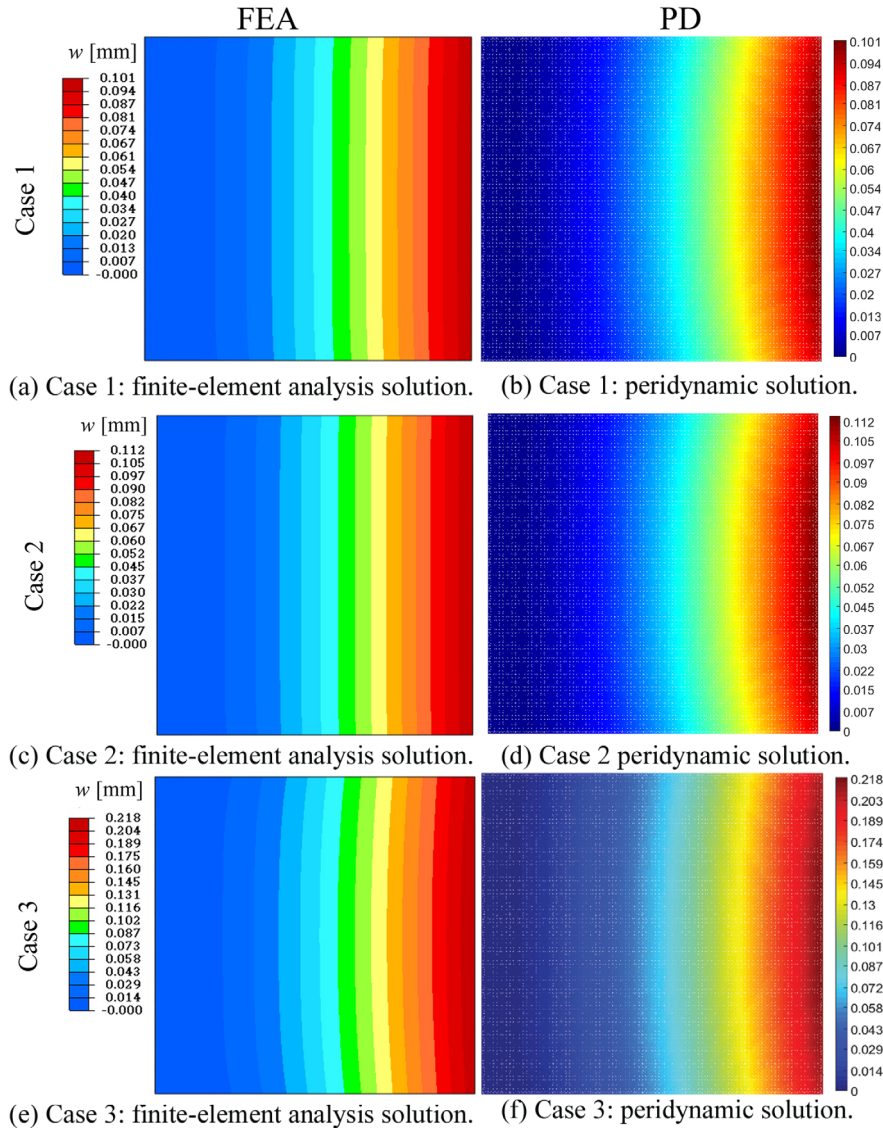


Figure 8. Finite-element analysis (FEA) and peridynamic (PD) comparison of the out-of-plane displacement contours of the orthotropic plates subjected to transverse loading.

subjected to the body forces $\hat{\mathbf{b}} = 5 \times 10^8 \text{ N/m}^2 \mathbf{k}$ and $\hat{\mathbf{b}} = 1 \times 10^8 \text{ N/m}^2 \mathbf{k}$, respectively. Damage plots of 0° and 90° plates with pre-existing central cracks are shown in Figure 16. The directions of crack propagation are indicated by the damage plots. The crack within the 0° plate propagates in the same direction as the existing crack and the crack within the 90° plate propagates perpendicular to the existing crack. The crack propagation paths for both 0° and 90° oriented plates follow the orthotropic properties of the materials, as expected.

The reason for asymmetry in crack propagation in the 0° oriented orthotropic plate comes from the definition of the failure criteria, which depends on the transverse shear angles, as shown in equation (84). Since the transverse shear angle is directly related to the out-of-plane deformation, w , and since the deformation ranges from 0 at the left boundary to the maximum at the right boundary, the transverse shear angle also increases from the left edge to the right edge. Therefore, it is expected that the crack growth would not be symmetric. To obtain a symmetric crack growth, we modeled a 0° oriented orthotropic plate under a uniform pressure of $\hat{\mathbf{b}} = 1 \times 10^7 \text{ N/m}^2 \mathbf{k}$ with both left and right edges clamped instead of one edge, as illustrated in Figure 17. Since the out-of-plane deformation is now symmetric,

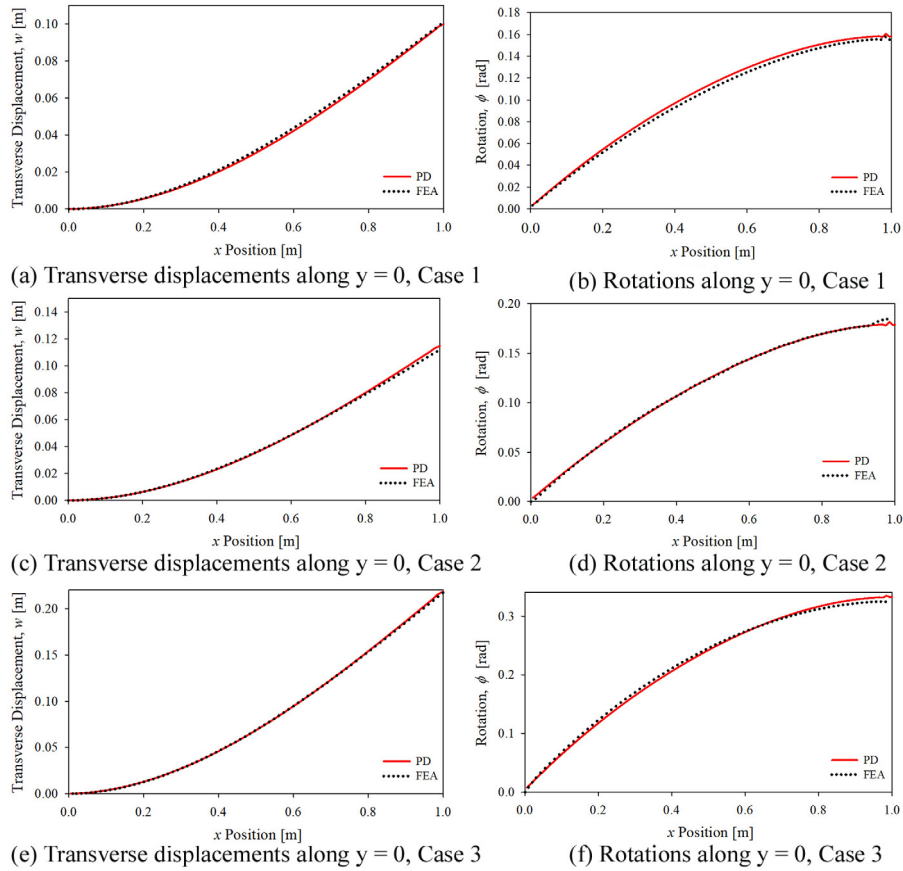


Figure 9. Displacements and rotations along $y = 0$ line of orthotropic plates subjected to transverse loading. FEA: finite-element analysis; PD: peridynamic.

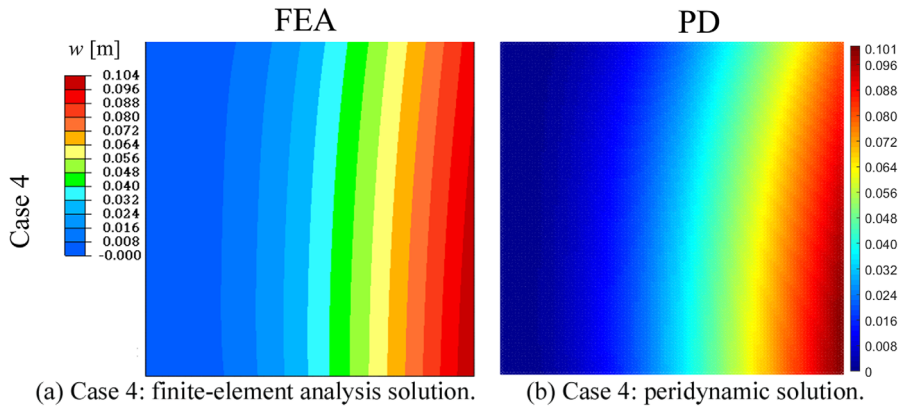


Figure 10. Comparison of out-of-plane displacement contours of 45° oriented plate subjected to transverse loading: (a) finite-element analysis (FEA); (b) peridynamic (PD) theory.

the transverse shear deformation is also symmetric. The symmetric crack propagation is depicted in Figure 18.

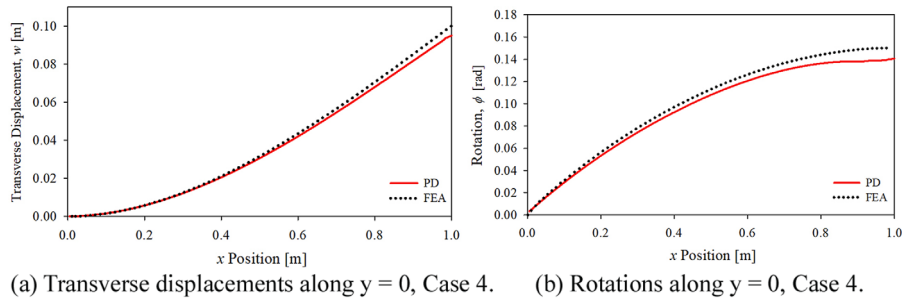


Figure 11. Displacements and rotations along $y = 0$ line of 45° oriented plate subjected to transverse loading. FEA: finite-element analysis; PD: peridynamic.

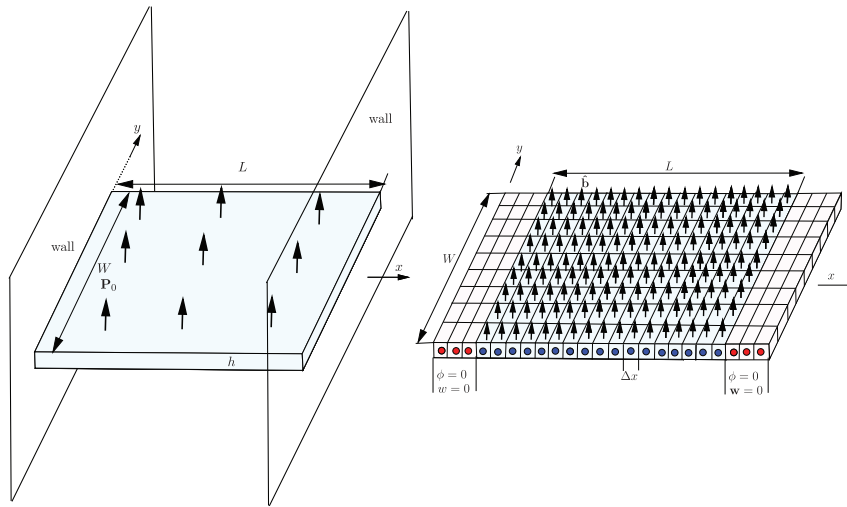


Figure 12. Mindlin plate subjected to uniform pressure.

9. Conclusions

In this study, a new peridynamic plate formulation for orthotropic thick plates under bending loads is presented. The peridynamic equations of motion for orthotropic thick plates are obtained by applying the principal of virtual work. A new bond strength criterion, based on the critical curvatures and the critical transverse shear angle, is also proposed. Peridynamic analyses are conducted using the numerical solution with the ADR algorithm in the developed MATLAB code. First, an isotropic elastic solution of a square plate subjected to transverse loading is performed. The numerical results show that the orthotropic formulation transforms into the isotropic plate formulation previously developed by Diyaroglu et al. [46]. The results obtained from this study are also compared with three-dimensional FEA solutions in ABAQUS. For orthotropic materials, nine different cases are solved by changing the material properties, material orientations, and boundary conditions. Six of the cases are solved for a one-edge clamped boundary condition and the other three cases are solved for the condition where both edges are clamped. By comparing the obtained peridynamic results with those obtained using FEA, it can be concluded that the proposed peridynamic Mindlin formulation for orthotropic plates is valid and can capture out-of-plane deflections and rotations of the orthotropic plates subjected to transverse loads. After validation of the elastic results, crack propagation in orthotropic plates with central cracks under bending loading is studied. The analyses are performed for plates oriented at 0° and 90° under transverse loading and for a plate oriented at 0° under uniform pressure.

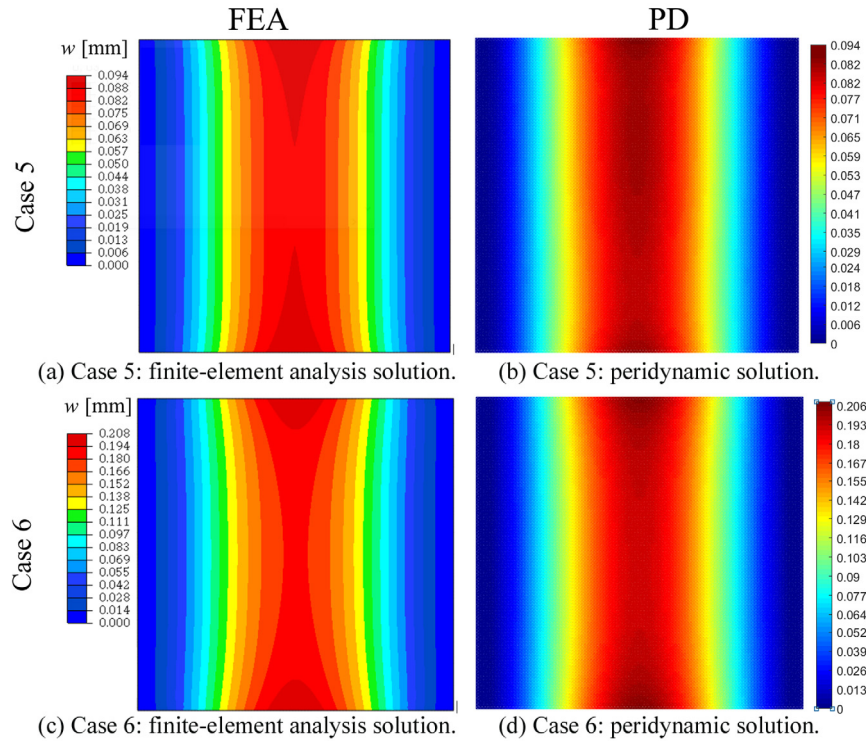


Figure 13. Finite-element analysis (FEA) and peridynamic (PD) comparison of the out-of-plane displacement contours of orthotropic plates subjected to uniform pressure.

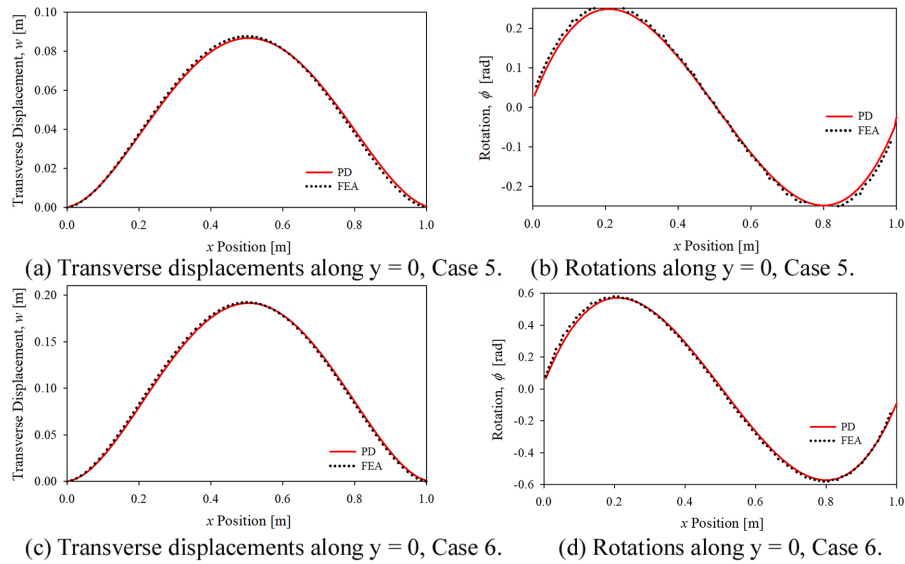


Figure 14. Displacements and rotations along $y = 0$ line of orthotropic plates subjected to uniform pressure. FEA: finite-element analysis; PD: peridynamic.

Table 2. Plate material configurations for Mindlin plates under uniform pressure.

Case	E_1 (GPa)	E_2 (GPa)	G (GPa)
5	200	20	75
6	90	50	33.75

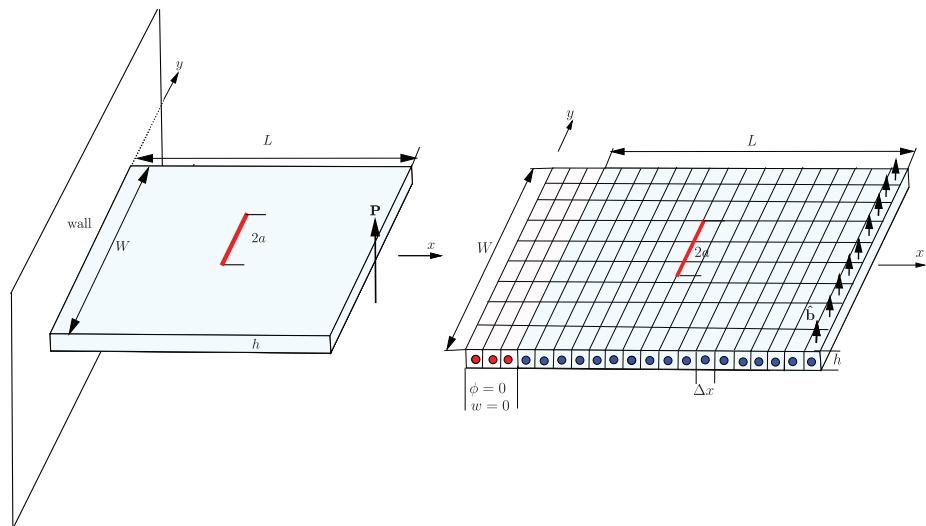


Figure 15. Mindlin plate with pre-existing crack subjected to transverse force.

Table 3. Plate material parameters for Mindlin plates with central cracks subjected to transverse load.

E_1 (GPa)	E_2 (GPa)	G (GPa)
300	16	60

Table 4. Fracture toughness of Mindlin plates with central crack subjected to a transverse load.

G_I^{fiber} (MPa.m)	G_I^{matrix} (MPa.m)	G_{III} (MPa.m)
4.96×10^{-4}	4.51×10^{-3}	1.88×10^{-3}

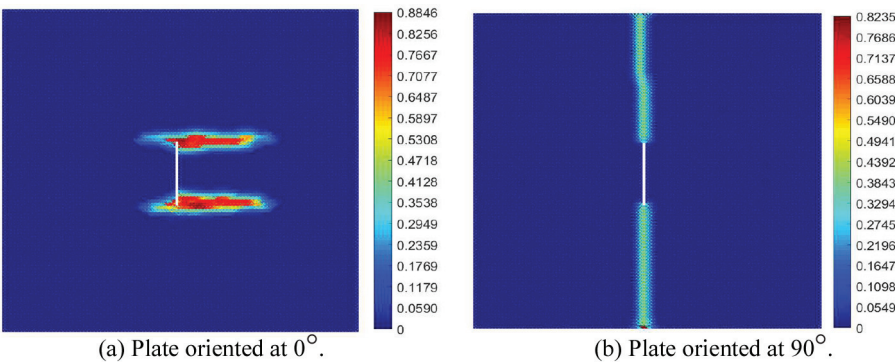


Figure 16. Damage contours of plates oriented at (a) 0° and (b) 90° subjected to transverse force.

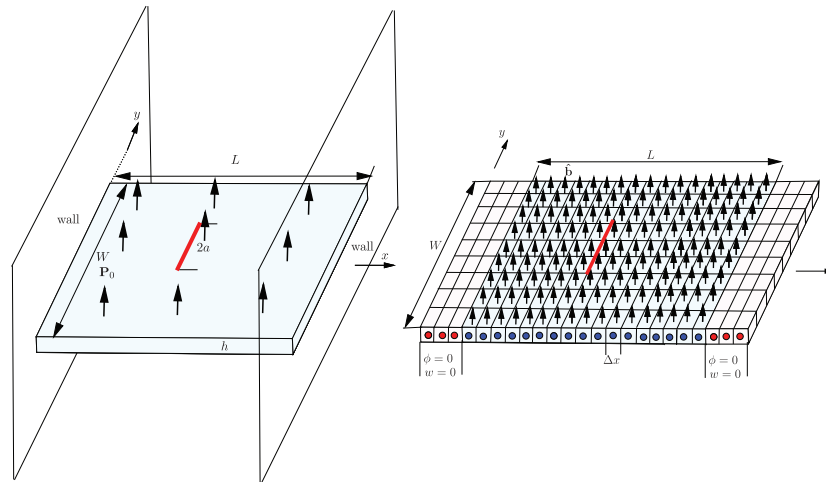


Figure 17. Mindlin plate with a pre-existing crack subjected to uniform pressure.

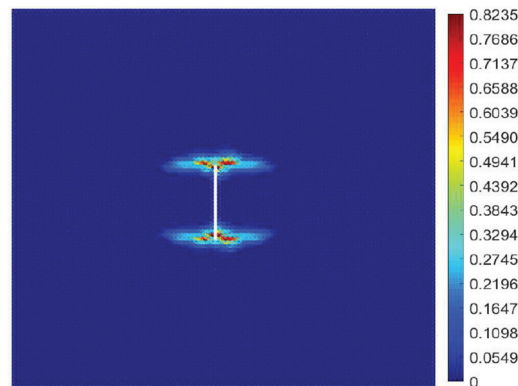


Figure 18. Damage contours of plate oriented at 0° subjected to uniform pressure.

In summary, the proposed peridynamic formulation of an orthotropic Mindlin plate is able to capture the bending behavior of orthotropic plates. The proposed model can utilize the critical energy release rates and allow crack propagation. The crack propagation paths are also well predicted in numerical simulations.

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